

Understanding UDDI

(Universal Description, Discovery, and Integration)

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- 3 UDDI Use Cases and Benefits/Challenges

What is UDDI?

- UDDI stands for Universal Description, Discovery, and Integration.
- UDDI is a specification for distributed registry of web services.
- Platform-independent open framework for publishing and discovering services over the internet.
- Part of the Web Services stack alongside SOAP, WSDL, etc.

History of UDDI

- Developed by IBM, Microsoft, and Ariba in 2000.
- Part of the larger SOA (Service-Oriented Architecture) ecosystem.
- Key Milestones:
 - UDDI v1 released in 2000.
 - UDDI v2 released in 2001.
 - UDDI v3 released in 2003.

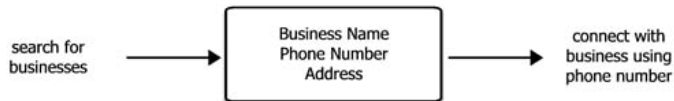
Importance of UDDI

- Provides a **registry** for web service descriptions.
- Facilitates discovery of services for integration.
- Promotes reusability and standardization of services.

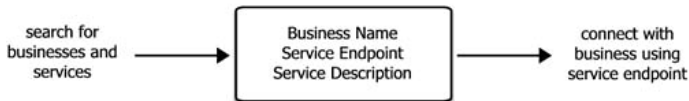
UDDI & Web Services

- Web services are self-contained applications accessible over internet.
- UDDI facilitates the discovery and description of these services.
- Supports SOAP and WSDL standards for service communication.

Understanding UDDI with Analogy



(a) Telephone Directory Book



(b) UDDI Registry

- Key Components:

- UDDI Registry: A directory where businesses register their web services.
- White Pages: Contact information and basic data.
- Yellow Pages: Service categories and classifications.
- Green Pages: Technical details about how to invoke services (WSDL, bindings).

UDDI Registry Explained

- White Pages: Business and contact information.
- Yellow Pages: Categorizes services based on taxonomies (NAICS, UNSPSC, etc.).
- Green Pages: Technical descriptions for using services, including URLs, protocols, and bindings.

UDDI registry can be either:

- **Public Registries:** Accessible by anyone to publish & find services, often used by businesses to share info. with partners and clients.
- **Private Registries:** Deployed within organizations for internal service discovery and integration.

White Pages (BusinessEntity)

- Provides basic business information such as name, description, and contact details.
- **Purpose:** It identifies & describes the organization providing service.
- Contents:
 - Business name and description.
 - Contact information (address, phone, email).
 - Unique identifiers (like D-U-N-S¹ numbers for business verification).
- Acts as an identity card for businesses listed in the UDDI registry.

¹A D-U-N-S Number is a 9-digit unique identifier for businesses. Live Business Identity, which is derived from the Dun & Bradstreet Data Cloud and aggregates all the information provided for any given entity within the Data Cloud.

Yellow Pages (BusinessService)

- **Purpose:**
Categorizes services based on function, industry, or service type.
- Yellow Pages classify web services offered by businesses using various taxonomies and classification schemes.
- Uses taxonomies like:
 - **NAICS (North American Industry Classification System).**
 - **UNSPSC (United Nations Standard Products & Services Code).**
 - Custom taxonomies can also be used.
- Allows services to be easily discovered by category.

Green Pages (BindingTemplate)

- **Purpose:**
To provide concrete information about how to connect to the services, such as endpoints, protocols, and access methods.
- Contains technical details necessary to access and invoke web services listed in the UDDI registry.
- Contents:
 - Service endpoints (URLs).
 - Protocols used (SOAP, REST).
 - Service binding information (WSDL documents).
 - Invocation details, including security policies and input/output formats.
- Provides the technical blueprint for invoking services.

- Core Data Structures:
 - BusinessEntity: Represents a company.
(White Pages)
 - BusinessService: Represents a service provided by a company.
(Yellow Pages)
 - BindingTemplate: Describes how to invoke a service.
(Green Pages)
 - tModel: Defines service standards.

The Global Directory: UDDI Architecture



White Pages (BusinessEntity)

- Business Name & Description
- Contact Information
- Unique identifiers (D-U-N-S)



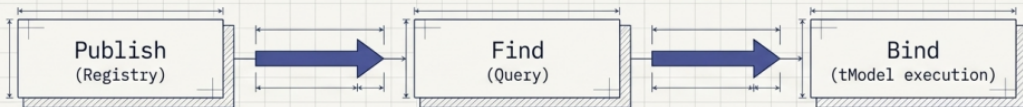
Yellow Pages (BusinessService)

- Service Categorization
- Industry Taxonomies
- NAICS & UNSPSC codes

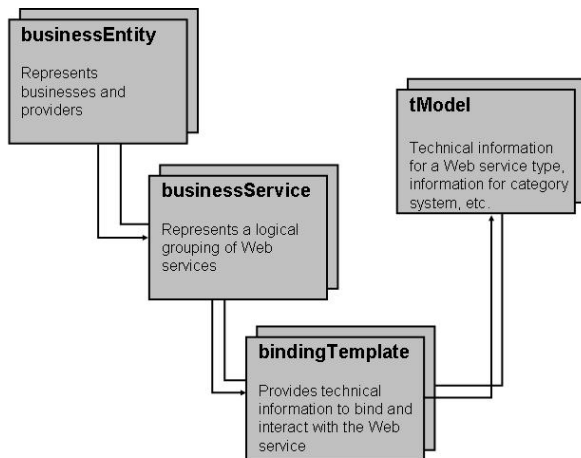


Green Pages (BindingTemplate)

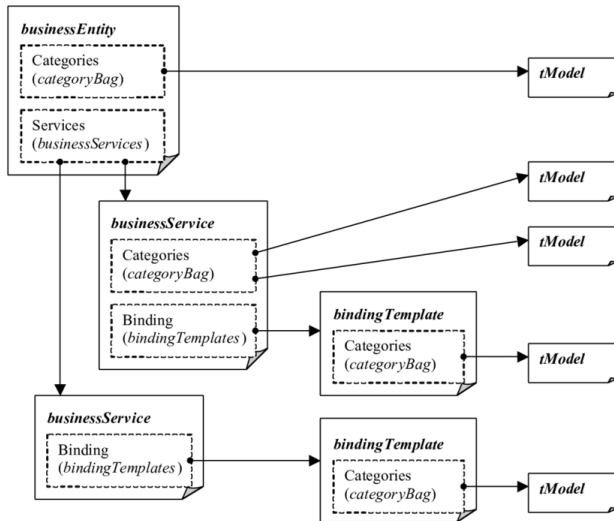
- Service endpoints (URLs)
- Protocols (SOAP/REST)
- WSDL bindings & tModel



UDDI Data Model



UDDI Data Model ... Elaborated



tModel (Technical Model)

- Represents a reusable service type or technical specification.
- **Purpose:**
 - Defines service standards and protocols.
 - Ensures interoperability between different systems.
- Describes how services conform to certain protocols or standards.
- Contents:
 - Technical standards (SOAP, WSDL, HTTP, etc.).
 - Service interfaces and contract definitions.

UDDI Publishing Process

- Business publishes details (BusinessEntity).
- Business publishes services (BusinessService).
- Provide binding information (BindingTemplate).
- Define the technical details (tModel).

UDDI Discovery Process

- Query the UDDI registry.
- Search by business name, service type, or tModel.
- Retrieve service descriptions and bindings.
- Invoke the service using details in the Green Pages.

	<i>Business</i>	<i>Service</i>	<i>Binding</i>	<i>tModel</i>
Save/Update	save_business	save_service	save_binding	save_tModel
Delete	delete_business	delete_service	delete_binding	delete_tModel
Find	find_business	find_service	find_binding	find_tModel
GetDetail	get_businessDetail	get_serviceDetail	get_bindingDetail	get_tModelDetail

Figure: CRUD API messages defined by the UDDI Specs.

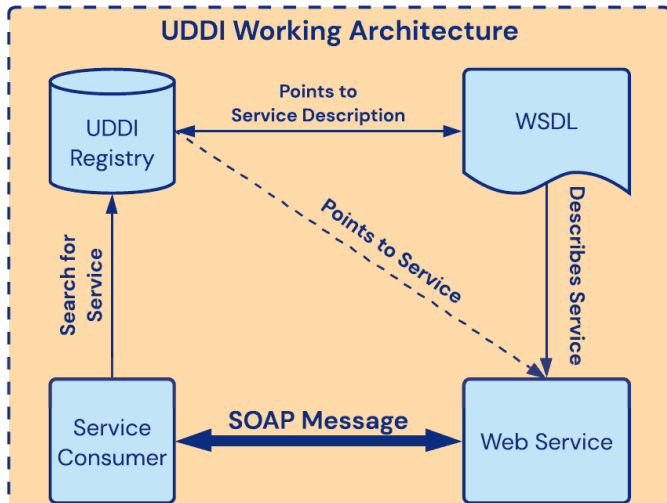
UDDI API Interactions

- Key UDDI APIs:
 - Inquiry API: For searching services in the registry.
 - Publish API: For publishing business and service information.
 - Security API: For managing access and authentication.
 - Subscription API: For monitoring changes in the registry.

	<i>Business</i>	<i>Service</i>	<i>Binding</i>	<i>tModel</i>
Save/Update	save_business	save_service	save_binding	save_tModel
Delete	delete_business	delete_service	delete_binding	delete_tModel
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Figure: CRUD API messages defined by the UDDI Specs.

UDDI Schematic Model



- Service discovery in SOA environments.
- Facilitates B2B integrations.
- Companies use UDDI to publish and access third-party services.
- Examples: Cloud service discovery, payment gateways.

UDDI: Benefits and Challenges

Benefits of UDDI

- Standardized approach to service discovery.
- Promotes interoperability and reuse of web services.
- Simplifies service integration in large, distributed systems.
- Reduces the time and effort to find services.

Challenges with UDDI

- Initially slow adoption due to complexity.
- Decline in use with the rise of REST APIs and modern alternatives.
- Security concerns with open registries.
- Competing discovery methods like DNS-based services.

Conclusion

- UDDI plays a critical role in service-oriented architectures by enabling service discovery.
- It was foundational in shaping web service interactions.
- Legacy systems still utilize UDDI for service registry and discovery.